

Experimental context and variability in 337 LGBM-matched HuSkinDB measurements

Technical analysis of conditions, references, and within-compound variability for the LGBM ± 0.6 subset.

Technical summary

The 337 selected measurements represent **171 compounds** and **71 reference/DOI pairs**. Only 54 compounds have repeated selected measurements; 117 compounds therefore have unknown within-compound variability. Among repeated compounds, 21 span more than 0.5 log10 unit and three exceed 1.0 log10 unit. Their median experimental range is 0.381 log10 unit, versus a median LGBM prediction range of 0.0137. The selected data describe substantial protocol/source heterogeneity that the current model largely cannot vary with.

Source: HuSkinDB LGBM ± 0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Selected measurements

337

Measurement rows satisfying the inclusive ± 0.6 absolute-error rule.

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Table:
lgbm_within_0.6_selected_measureme
nt_conditions.csv

Unique compounds

171

Distinct compounds identified by canonical SMILES.

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Repeated compounds

54

Compounds with at least two selected measurement rows.

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nt_conditions.csv

References

71

Distinct reference and DOI pairs represented.

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Condition cells available (%)

70.79

Available values across 337 rows and 14 experimental-condition fields.

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Experimental variability is concentrated in a minority of repeat-measured compounds

Estradiol, 1-butanol, and caffeine have selected-subset experimental ranges above 1.0 log₁₀ unit; methotrexate reaches exactly 1.000. For the 54 repeated compounds, the median range is 0.381 and 22 receive an identical LGBM prediction across all their selected rows. Even the largest prediction range is below 0.1. This gap indicates limited representation of row-specific experimental context, not proof that any one condition caused the observed differences.

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Largest within-compound experimental logK_p ranges

Source: HuSkinDB LGBM ± 0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Compound	Experimental logK _p range
Estradiol	1.13log ₁₀ (cm/s)
1-Butanol	1.08log ₁₀ (cm/s)
Caffeine	1.02log ₁₀ (cm/s)
Methotrexate	1log ₁₀ (cm/s)
1-Hexanol	0.92log ₁₀ (cm/s)
1-Decanol	0.91log ₁₀ (cm/s)
Fentanyl	0.84log ₁₀ (cm/s)

Compound	Experimental logKp range
Ethanol	0.8log10(cm/s)
Salicylic acid	0.8log10(cm/s)
Nicotine	0.77log10(cm/s)
Sufentanil	0.75log10(cm/s)
Testosterone	0.74log10(cm/s)
Bis(2-ethylhexyl)phthalate	0.67log10(cm/s)
1-Pentanol	0.6log10(cm/s)
1-Octanol	0.56log10(cm/s)

The table pairs each compound's minimum and maximum selected experimental rows. Read the listed condition differences as a protocol audit: several fields usually change together, so the table cannot assign the range to a single factor.

Minimum-versus-maximum rows for the most variable compounds

Source: HuSkinDB LGBM ± 0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Compound	Rows	Min logKp	Max logKp	Range	Fold range	Predictio n range	Minimum reference	Maximum reference	Reported fields that differ
Estradiol	12	-6.68	-5.56	1.13	13.43	0.03	Goodman, 1988	Pershing, 1993	skin source type used layer skin preparation storage temperature (°C) storage duration (days) donor/skin surface temperature (°C) donor type acceptor type cell type
1-Butanol	10	-6.56	-5.48	1.08	11.89	0.03	Scheuplein, 1973	Chantasart, 2010	skin source type used layer storage temperature (°C) neat donor/skin surface temperature (°C) acceptor temperature (°C) acceptor type cell type
Caffeine	6	-7.56	-6.54	1.02	10.35	0.04	Mitragotri, 1995	Wilkinson, 2006	skin source type skin source site used layer skin preparation storage temperature (°C) storage duration (days) donor/skin surface temperature (°C) donor type acceptor temperature (°C) acceptor pH acceptor type cell type
Methotrexate	7	-7.5	-6.5	1	10	0	Vaidyanathan, 1985	Vaidyanathan, 1985	donor pH
1-Hexanol	10	-5.78	-4.86	0.92	8.3	0.08	Cross, 2003	Bond, 1988	skin source type skin source site skin preparation storage temperature (°C) storage duration (days) donor/skin surface temperature (°C) donor pH donor type acceptor temperature (°C) acceptor pH acceptor type cell type
1-Decanol	4	-5.52	-4.6	0.91	8.18	0.08	Buist, 2010	Mitragotri, 2000	skin source site skin preparation storage temperature (°C) donor pH donor type acceptor pH acceptor type cell type
Fentanyl	10	-5.85	-5.01	0.84	6.85	0.01	Roy, 1990	Roy, 1990	donor pH donor type
Ethanol	7	-7.08	-6.28	0.8	6.32	0.07	Blank, 1964	Cross, 2003	skin source type used layer skin preparation donor/skin surface temperature (°C) donor pH donor type acceptor temperature (°C) acceptor pH acceptor type cell type
Salicylic acid	3	-6.22	-5.42	0.8	6.31	0.02	Schmook, 2001	Degim, 1998	skin source type skin preparation storage temperature (°C) storage duration (days) donor/skin surface temperature (°C) donor type acceptor temperature (°C) acceptor pH acceptor type
Nicotine	3	-6.04	-5.27	0.77	5.83	0.01	Degim, 1998	Hadgraft, 1987	skin source type skin preparation storage temperature (°C) storage duration (days) donor/skin surface temperature (°C) donor pH donor type acceptor pH acceptor type
Sufentanil	9	-5.76	-5.02	0.75	5.57	0	Roy, 1990	Roy, 1990	donor pH donor type
Testosterone	11	-6.21	-5.48	0.74	5.47	0.02	Mitragotri, 1995	Sznitowska, 2001	skin source site used layer skin preparation storage temperature (°C) storage duration (days) donor/skin surface temperature (°C) donor pH acceptor temperature (°C) acceptor type cell type

Compound	Rows	Min logKp	Max logKp	Range	Fold range	Prediction range	Minimum reference	Maximum reference	Reported fields that differ
Bis(2-ethylhexyl)phthalate	2	-7.65	-6.98	0.67	4.72	0	Buist, 2010	Barber, 1992	skin source type skin source site used layer skin preparation storage temperature (°C) storage duration (days) neat donor type acceptor temperature (°C) acceptor pH acceptor type cell type
1-Pentanol	4	-5.78	-5.18	0.6	4	0.08	Blank, 1964	Scheuplein, 1973	used layer skin preparation storage temperature (°C) donor/skin surface temperature (°C) donor type acceptor temperature (°C) acceptor type cell type
1-Octanol	9	-5.14	-4.59	0.56	3.59	0.06	Scheuplein, 1973	Cross, 2003	skin source type used layer storage temperature (°C) donor/skin surface temperature (°C) donor pH donor type acceptor temperature (°C) acceptor pH acceptor type cell type

Missing experimental context limits adjustment for protocol effects

Across 4,718 condition cells, 70.8% are available. The median measurement is missing four of 14 condition fields; only 13 rows report all 14. Storage duration is unavailable in 71.5% of rows, acceptor pH in 58.2%, and donor pH in 54.0%. Used layer is nearly complete (99.4%) and the neat indicator is complete.

Source: HuSkinDB LGBM ±0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Unavailable experimental-condition fields

Source: HuSkinDB LGBM ±0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Condition field	Unavailable rows
storage duration (days)	71.51%
acceptor pH	58.16%
donor pH	54.01%
donor type	37.09%
acceptor type	35.61%
storage temperature (°C)	33.53%
donor/skin surface temperature (°C)	30.27%
acceptor temperature (°C)	26.11%
cell type	21.36%
skin preparation	15.73%

Condition field	Unavailable rows
skin source type	12.76%
skin source site	12.17%
used layer	0.59%
neat	0%

Protocol signatures are completely coupled to literature sources

The citations span 1964–2012. Coverage is not dominated by one paper: the top five references contribute 22.3% of rows and the top ten contribute 35.9%. However, all 140 recorded condition signatures occur within only one reference. Consequently, reference/laboratory effects and protocol effects cannot be separated using simple grouped comparisons in this selected subset.

Source: HuSkinDB LGBM ±0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Largest references in the selected subset

Source: HuSkinDB LGBM ±0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Reference	Selected measurement rows
Roberts, 1977	18
Scheuplein, 1973	17
Roy, 1990	16
Buist, 2010	14
Zhang, 2009	10
Chantasart, 2010	10
Sznitowska, 2001	10
Anderson, 1988	9
Cross, 2003	9
Mitragotri, 2000	8
Scheuplein, 1965	8
Swart, 2005	8
Xu, 2002	8
Michaels A. S., 1975	8

Hadgraft, 1987

7

Same-reference pH series are the clearest condition-linked examples

Within one citation, methotrexate spans 1.000 log10 unit while reported donor pH changes. The Roy (1990) selected series spans 0.836 for fentanyl and 0.746 for sufentanil, with strong monotonic rank associations across reported pH (Spearman $\rho \approx 0.98$), although one pH 7.4 row also differs in skin site, layer, and preparation. These are biologically plausible condition-linked patterns, but they remain observational and partially confounded.

Source: HuSkinDB LGBM ± 0.6 selected measurement table

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Experimental logKp across reported donor-pH series

Source: HuSkinDB LGBM ± 0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Reported donor pH	Experimental logKp	Compound
6.02	-5.85log10(cm/s)	Fentanyl
6.95	-5.71log10(cm/s)	Fentanyl
7.4	-5.5log10(cm/s)	Fentanyl
7.43	-5.45log10(cm/s)	Fentanyl
7.95	-5.21log10(cm/s)	Fentanyl
8.52	-5.12log10(cm/s)	Fentanyl
9.04	-5.01log10(cm/s)	Fentanyl
9.37	-5.04log10(cm/s)	Fentanyl
1.98	-7.5log10(cm/s)	Methotrexate
2.98	-7.21log10(cm/s)	Methotrexate
3.87	-6.61log10(cm/s)	Methotrexate
4.12	-7.01log10(cm/s)	Methotrexate
4.12	-6.81log10(cm/s)	Methotrexate
5.29	-6.5log10(cm/s)	Methotrexate
6.34	-6.84log10(cm/s)	Methotrexate
6.02	-5.76log10(cm/s)	Sufentanil
6.95	-5.55log10(cm/s)	Sufentanil

Reported donor pH	Experimental logKp	Compound
7.4	-5.37log10(cm/s)	Sufentanil
7.43	-5.36log10(cm/s)	Sufentanil
7.95	-5.19log10(cm/s)	Sufentanil
8.52	-5.08log10(cm/s)	Sufentanil
9.04	-5.02log10(cm/s)	Sufentanil
9.37	-5.03log10(cm/s)	Sufentanil

Repeated compound series within a single reference

Source: HuSkinDB LGBM ± 0.6 selected measurement table

Table: lgbm_within_0.6_selected_measurement_conditions.csv

Compound	Reference	Rows	Min logKp	Max logKp	Range	Varying reported conditions	Donor-pH Spearman ρ	p-value
Methotrexate	Vaidyanathan, 1985	7	-7.5	-6.5	1	donor pH	0.61	0.14
Estradiol	Pershing, 1993	2	-6.56	-5.56	1	donor type	—	—
Fentanyl	Roy, 1990	8	-5.85	-5.01	0.84	skin source site used layer skin preparation donor pH donor type	0.98	0
Sufentanil	Roy, 1990	8	-5.76	-5.02	0.75	skin source site used layer skin preparation donor pH donor type	0.98	0
1-Hexanol	Cross, 2003	3	-5.78	-5.05	0.73	used layer skin preparation	—	—
Estradiol	Goodman, 1988	2	-6.68	-6.02	0.67	donor type acceptor type	—	—
1-Pentanol	Scheuplein, 1973	2	-5.78	-5.18	0.6	used layer	—	—
Ephedrine	Michaels A. S., 1975	4	-6.33	-5.78	0.55	donor pH	0.77	0.23
Lysine	Sznitowska, 1993	2	-7.33	-6.86	0.47	donor pH	—	—
1-Butanol	Scheuplein, 1973	2	-6.56	-6.16	0.4	used layer neat	—	—
Butoxyethanol	Wilkinson, 2006	2	-6.36	-5.98	0.38	skin preparation	—	—
2-Propoxyethanol	Venier, 2004	2	-6.92	-6.54	0.38	neat	—	—
Caffeine	Wilkinson, 2006	2	-6.92	-6.54	0.37	skin preparation	—	—
Ethylene glycol mono methyl ether acetate	Venier, 2004	2	-6.64	-6.3	0.34	neat	—	—
Hydrocortisone	Sznitowska, 2001	5	-7.68	-7.36	0.32	donor pH donor type	-0.87	0.05

Scope, data, and definitions

One row is one reported measurement. Compounds are grouped by canonical SMILES. The cohort includes only rows with absolute LGBM prediction error ≤ 0.6 . Experimental variability is max minus min logKp within a compound; fold range is 10 raised to that difference. Unknown, empty, N/A, and not-reported values count as unavailable. A condition context combines all 14 condition fields with reference and DOI.

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Methodology

The analysis profiles completeness, citation concentration, repeated-measurement counts, within-compound experimental and prediction ranges, condition-signature uniqueness, and repeated series within a citation. It preserves every source measurement and adds annotations rather than averaging measurements. Exact row counts, calculations, and charts are reproducible in the companion notebook.

Limitations and robustness checks

This subset is selected using experimental logKp, creating outcome-dependent selection bias; it is not an unbiased validation cohort. Most compounds have only one selected measurement. Conditions are incompletely reported, several fields co-vary, and condition signatures are nested within references. The pH associations are descriptive, not causal. Row count, source-row uniqueness, the inclusive ± 0.6 rule, 171-compound count, 71-reference count, and notebook execution were verified.

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Recommended next steps

1. Fit any condition-effect analysis on the full, unfiltered HuSkinDB table, not this outcome-selected subset.
2. Use a hierarchical model with compound and reference effects before estimating pH, temperature, layer, preparation, donor/acceptor medium, or cell-type associations.
3. Add ionization/pH and other protocol variables to future prediction models; the current LGBM condition input is limited to experimental temperature ($T_{exp,i}$).
4. Treat single-measurement compounds as variability unknown, not variability low.

Further questions

Can the original papers resolve the high missingness in storage duration and pH? Are units and skin-thickness conventions harmonized across references? Does a full-dataset multilevel model retain the pH patterns after controlling for compound, reference, skin layer, preparation, and medium?